Wenhao Peng

RF Acoustic Wave Technology

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Biography

Wenhao Peng received the Ph.D. degree in electrical and computer engineering and the M.S.E. degree in mechanical engineering from the University of Michigan, Ann Arbor, MI, USA, in 2026 and 2024, respectively. He received the M.S. degree in electrical and computer engineering from the University of Michigan in 2019 and the B.S.E. degrees from the University of Michigan and Shanghai Jiao Tong University, Shanghai, China, in 2018. His published modeling work connects thermodynamics and mechanical vibrations with resonator and filter design. In ferroelectric BAW technology, he developed a physics based equivalent circuit model that bridges nonlinear ferroelectric thermodynamics and resonator DC bias dependent response. In piezoelectric BAW technology, he derived an energy based expression for electromechanical coupling as the ratio of converted electrical energy to supplied elastic energy and proved by modal analysis its consistency with the coefficient determined from the series and parallel resonance frequencies when the mode is sufficiently isolated from other modes, making modal field distributions useful for evaluating layer stacks and device performance at mm-Waves. Experimentally, he demonstrated the first periodically poled AlN/ScAlN/AlN trilayer FBAR operating above 50 GHz, a quad band switchless filter bank, an acoustic phase inversion switch, and an intrinsically switchable balanced filter. He has published in IEEE T-MTT, IEEE MWTL, IEEE MWCL, Applied Physics Letters, has presented as the first author at IEEE/MTT-S International Microwave Symposium in 2021, 2024, and 2025, and serves as a peer reviewer for leading journals in IEEE, APS, and AIP.

Original Contributions

Modal analysis for resonator linear response	Derived an energy based expression for the electromechanical coupling coefficient (k_t^2) as the ratio of converted electrical energy to supplied elastic energy, and proved by modal analysis its consistency with the coefficient determined from the series and parallel resonance frequencies. The result makes acoustic fields at resonance quantitatively interpretable. It shows the contribution of each layer to the energy terms in resonator coupling, enables direct evaluation of stack performance at mm-Waves, and reduces reliance on iterative optimization against simulated admittance response. This relation was presented with its derivation at IMS 2024 and IMS 2025.
Analytical dynamics for resonator nonlinearity	Developed the first equivalent circuit model of ferroelectric driven BAW resonators based on thermodynamics of ferroelectric materials by utilizing acoustic electromagnetic duality and analytical dynamics. The model is validated with measured response across DC bias. First author of the corresponding IEEE T-MTT 2022 and IMS 2021 papers.
mm-Wave scaling by periodically poled films	Demonstrated the first periodically poled AlN/ScAlN/AlN trilayer FBAR operating at 56 GHz along with the modal analysis results, which guided electrode and layer stack selection for a second generation 36 GHz device. First author of the corresponding IMS 2024 and IMS 2025 papers.
Switchless RF frontends	Demonstrated the first switchless quad band filter bank based on ferroelectric BST FBARs, the first intrinsically switchable balanced ferroelectric FBAR filter, and the first RF acoustic phase inversion switch based on poled ferroelectrics.

Experience

Feb 2026–Present **Staff Electrical Engineer**, *Skyworks Solutions Inc*, Irvine, CA.

- Provide device modeling and simulation support for BAW and SAW filter designs.
- Participate in and develop the filter design workflow.
- Diagnose resonator and filter performance issues through mass production using equivalent circuit and multiphysics models.

Sep 2019–Dec 2025 **Graduate Student Research Assistant**, *University of Michigan*, Ann Arbor, MI.

Advisor: Prof. Amir Mortazawi

mm-Wave FBARs based on periodically poled III-nitride thin films

- Demonstrated the first periodically poled AlN/ScAlN/AlN trilayer FBARs at 56 GHz, followed by a second generation 36 GHz resonator with a redesigned electrode and layer stack.
- Derived an energy based expression for electromechanical coupling from the open circuit eigenmodes of an FBAR. It evaluates the coupling coefficient as the ratio of converted electrical energy to supplied elastic energy, which is proved by modal analysis to be consistent with the value determined from the series and parallel resonance frequencies. The expression enabled direct evaluation of layer contributions to the energy terms and guided electrode and layer stack selection for the mm-Wave FBAR work.
- Advanced the nanofabrication and pulsed-field polarization switching processes of MBE grown AlN/ScAlN multilayer structures.
- Performed mm-Wave characterization up to 67 GHz with on-wafer Thu Reflect Line and open short de-embedding structures validated in ADS Momentum and Ansys HFSS.

Reconfigurable RF frontends based on intrinsically switchable BST FBARs

- Developed the first equivalent circuit model of ferroelectric driven BAW resonators based on ferroelectric thermodynamics. Applied analytical dynamics methods to bridge the material thermodynamic description and a compact circuit representation of acoustic resonance. Validated the model by direct comparison with measurements across DC bias.
- Demonstrated the first switchless quad band BST FBAR filter bank and the first intrinsically switchable balanced ferroelectric FBAR filter at 2 GHz.
- Demonstrated the first RF acoustic phase inversion switch based on poled ferroelectric stacked crystal filters.
- Synthesized filters using the image parameter method, applying Mason and mBVD resonator models, and completed cleanroom fabrication and DC bias dependent RF characterization.

Education

2026 **Ph.D., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.

Dissertation: *Layered Ferroelectric Millimeter Wave BAW Resonators: Modeling and Validation based on On-Wafer Characterization.* Advisor: Prof. Amir Mortazawi.

2024 **M.S.E., Mechanical Engineering**, *University of Michigan*, Ann Arbor, MI.

Dynamics and vibrations

2019 **M.S., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.

Integrated circuits and VLSI

2018 **B.S.E., Electrical Engineering**, *University of Michigan*, Ann Arbor, MI.

Summa Cum Laude

2018 **B.S.E., Electrical and Computer Engineering**, *Shanghai Jiao Tong University*, Shanghai, China.

Dual degree with U-M

Technical Skills

Functional materials	BAW resonator and filter design; thin film piezoelectrics and ferroelectrics (AlN, ScAlN, BST); periodically poled III-nitride multilayers; MEMS; RF frontend architectures
Modeling	Mason and mBVD equivalent circuits; thermodynamic modeling of ferroelectric nonlinearity; modal analysis; filter synthesis; multiphysics FEM (COMSOL); EM co-simulation (ADS Momentum, Ansys HFSS); analytical dynamics
Fabrication	Cleanroom process integration; photolithography; physical vapor deposition; wet/dry/plasma etching; thermal oxidation; device processing compatible with MBE-grown films; pulsed-field poling
Measurement	On-wafer S-parameter characterization up to 67 GHz; GSG probing; VNA calibration and de-embedding; DC bias dependent RF characterization; material parameter extraction from on-wafer characterization
Tools	ADS, COMSOL Multiphysics, Ansys HFSS, MATLAB, Python, Cadence
AI Aided Automation	LLM-assisted Python coding, custom multiphysics simulation tool development, reduced order finite element modeling

Journal Publications

2025 H. Desai, **W. Peng**, and A. Mortazawi, "Single-Pole Single-Throw RF Acoustic Phase Inversion Switch Leveraging Poled Ferroelectrics," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 1, pp. 6–13, 2025.

2023 D. Wang, P. Wang, S. Mondal, J. Liu, M. Hu, M. He, S. Nam, **W. Peng**, S. Yang, D. Wang, Y. Xiao, Y. Wu, A. Mortazawi, and Z. Mi, "Controlled Ferroelectric Switching in Ultrawide Bandgap AlN/ScAlN Multilayers," *Appl. Phys. Lett.*, vol. 123, no. 10, p. 103506, 2023.

2023 S. Nam, **W. Peng**, P. Wang, D. Wang, Z. Mi, and A. Mortazawi, "A mm-Wave Trilayer AlN/ScAlN/AlN Higher Order Mode FBAR," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 6, pp. 803–806, 2023.

2022 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Phenomenological Circuit Modeling of Ferroelectric-Driven Bulk Acoustic Wave Resonators," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 1, pp. 919–925, 2022.

2021 S. Nam, M. Z. Koohi, **W. Peng**, and A. Mortazawi, "A Switchless Quad Band Filter Bank Based on Ferroelectric BST FBARs," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 6, pp. 662–665, 2021.

Conference Publications

2025 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, "A 36 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR," *IEEE/MTT-S International Microwave Symposium (IMS)*, San Francisco, CA, 2025, pp. 786–789. *(presented by author)*

2024 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, "A 56 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR," *IEEE/MTT-S International Microwave Symposium (IMS)*, Washington, DC, 2024, pp. 150–153. *(presented by author)*

2021 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Physics Based Modeling of Electrostriction Based BAW Resonators," *IEEE MTT-S International Microwave Symposium (IMS)*, Atlanta, GA, 2021, pp. 214–217. *(presented by author)*

2020 M. Z. Koohi, **W. Peng**, and A. Mortazawi, "An Intrinsically Switchable Balanced Ferroelectric FBAR Filter at 2 GHz," *IEEE/MTT-S International Microwave Symposium (IMS)*, 2020, pp. 131–134.

2018 Y. Dai, **W. Peng**, Y. Wang, L.-X. Chuo, K. Suri, H. Zheng, D. Wentzloff, and H.-S. Kim, "Implementation and Evaluation of Bi-Directional WiFi Back-Channel Communication," *IEEE 29th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2018, pp. 1–7.

Peer Review Contributions

56 reviews **IEEE:** J. Microelectromechanical Systems (9), Microwave and Wireless Technology Letters (7), Electron Device Letters (4), J. Microwaves (2). **APS:** Physical Review Letters (5), Physical Review Applied (3), Physical Review Materials (2), Physical Review A (1), Physical Review B (2). **AIP:** Physics of Fluids (2). **Other:** Micromachines (11), Photonics (2), Sensors (2), Microwave (2), Applied Sciences (1), Progress in Quantum Electronics (1).

Selected Graduate Coursework

Wave Propagation in Elastic Solids; Advanced Vibrations of Structures; Advanced Engineering Acoustics; Theory of Solid Continua; Math Methods in ME; Advanced Solid State Microwave Circuits; Electromagnetics Theory; Introduction to MEMS; Monolithic Amplifier Circuits; Analog Integrated Circuits; VLSI Design; Computer Architecture; Digital Communication Signals and Systems; Probabilistic Methods in Engineering